

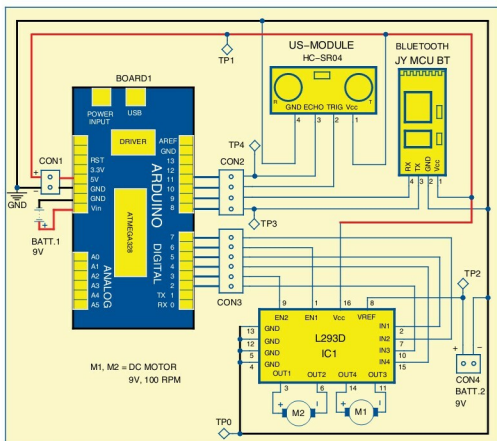


Fig. 3: Circuit of the Android phone-controlled robot

PARTS LIST

Semiconductors:	
IC1	- L293D motor driver
Miscellaneous:	
BOARD1	- Arduino UNO board
BLUETOOTH	- JY MCU BT Bluetooth module
US-module	- HC-SR04 ultrasonic transceiver module
CON1, CON4	- 2-pin connector
CON2	- 4-pin connector
CON3	- 6-pin connector
M1, M2	- 9V, 100rpm DC motor
BATT.1, BATT.2	- 9V battery

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	9V
TP3	Train of pulses
TP4	High pulse corresponding to the distance of obstacle

tooth module, respectively. Pins 10 and 11 are connected to trigger (TRIG) and echo (ECHO) pins of HC-SR04, respectively. Pins 2 through 7 of BOARD1 are the output pins, which are connected to IC1 for controlling the motors. Pins 2, 3 and 4 of BOARD1 are connected to IN3, EN2 and IN4 of IC1 to control motor M1, and pins 5, 6 and 7 are connected to IN1, EN1 and IN2 of IC1 to control motor M2. EN1 and EN2 are used to control the speeds of the motors.

The control commands for the ro-

bot are sent from the phone using the app shown in Fig. 2. You can send the commands either by touching on various options on the phone's screen or through speech commands. Corresponding to control commands you select on the app, the related data is sent through Bluetooth of the phone. Data transmitted by the phone is received by Bluetooth module at the robot end. The received data is fed to pin 8 of BOARD1. The microcontroller on BOARD1 processes the received data and drives motors accordingly.

The robot also sends back the status. In the app you can see label 'Robot Status,' below which you can see the direction in which robot is moving. The robot continuously sends back information about the distance from the closest obstacle to the phone, and if it reaches too close to an obstacle the robot stops automatically.

Software

The software for the robot is written in Arduino programming language. The Arduino UNO is programmed using Arduino IDE software. Atmega328 on Arduino UNO comes with a boot loader that allows you to upload new code to it without using an external hardware programmer. It uses STK500 protocol to communicate. You can bypass the boot loader and program the microcontroller through ICSP (in-circuit serial programming) header, but using boot loader programming is quick and easy. Select the correct board from 'Tools→Board' menu in Arduino IDE and burn the program (sketch) through standard USB port in the computer.

The procedure for installing Android app on the phone is as follows:

1. Download the app (Android_BOT.apk) from this month's EFY DVD (or eflymag.com) and copy it to your Android smartphone (or tablet).



Fig. 4: Bluetooth module

on the board is programmed using Arduino programming language using Arduino development environment.

Pins 8 and 9 of BOARD1 are connected to pins Tx and Rx of the Blue-